

1. Loop Complexity. Java

1. Single(int n). $O(n)$

$n = 1000$

c = 0	i = 0	i < n	i++	c++
1	0	true	1	1
2	1	true	2	2
3	2	true	3	3
4	3	true	4	4
...	...	...	...	...
1000	999	true	1000	1000
	1000	false		

5 time loop of for will continue each run.
 $n = 1000$ $n \leq 10000$ $n * 2$

2. nested(int n) $O(n^2)$

$n = 1000$

c = 0	i	i < 1000	i++	j	j < 1000	j++
1	0	true	1	0	true	1
2	1	true	2	1	true	2
3	2	true	3	2	true	3
4	3	true	4	3	true	4
...	...	...	...	...	...	...
1000	999	true	1000	999	true	1000
	1000	false				
1001	0	true	1	0	true	1
1002	1	true	2	1	true	2
1003	2	true	3	2	true	3
1004	3	true	4	3	true	4
...	...	...	...	...	...	...
2000	999	true	1000	999	true	1000
	1000	false				
...	...	...	...	...	...	...
1000	1000	false				

3. Triangle (int n)

n = 1000

C = 0	i	i < n	i++	j	j < i	j++	C++
	0	true	1	0	false		
1 < 1	1	true	2	0	true false	1	1
				1	false		
3 < 2	2	true	3	0	true	1	2
				1	true	2	3
				2	false		
6 < 3	3	true	4	0	true	1	4
				1	true	2	5
				2	true	3	6
				3	false		

499500 < 999	true	1000	0	true	1	4985001
			1	true	2	4985002
			2	true	3	4985003
			⋮			
			998	true	999	4995000
			999	false		

1000 false

||

4. halving (int n)

$n = 1000$

$c = 0$	$i = 1000^{(n)}$	$i > 1$	$i /= 2$	$c++$
1	← 1000	true	500	→ 1
2	← 500	true	250	→ 2
3	← 250	true	125	→ 3
4	← 125	true	62	→ 4
5	← 62	true	31	→ 5
6	← 31	true	15	→ 6
7	← 15	true	7	→ 7
8	← 7	true	3	→ 8
9	← 3	true	1	→ 9
1		false	— —	

5. $n \log N$ (int n)

$n = 1000$

$C=0$	i	$i < n$	$i++$	j	$j < n$	$j*=2$	$C++$
1	0	true	1	1	true	2	1
2				2	true	4	2
3				4	true	8	3
4				8	true	16	4
10				512	true	1024	10
				1024	False		

11	1	true	2	1	true	2	11
12				2	true	4	12
13				4	true	8	13
20				512	true	1024	20
				1024	False		

9991	999	true	1000	1	true	2	9991
9992				2	true	4	9992
9993				4	true	8	9993
10000				512	true	1024	10000
				1024	False		
1000		False					

6. Sequential (int n).

 $n = 1000$

C = 0	i	i < n	i++	C++	
1 ←	0	true	1	1	
2 ←	1	true	2	2	
3 ←	2	true	3	3	
4 ←	3	true	4	4	
5 ←	4	true	5	5	
	⋮				

1000 ← 999 true 1000 1000
 1000 false ——— 11

C = 1000	i	i < n	i++	C++
1001 ←	0	true	1	1001
1002 ←	1	true	2	1002
1003 ←	2	true	3	1003
1004 ←	3	true	4	1004
1005 ←	4	true	5	1005
1006 ←	5	true	6	1006
	⋮			

2000 ← 999 true 1000 2000
 1000 false ——— 11

1. sqrtLoop(int n)

$n = 1000$

C = 0	i	$i * i \leq n$	i++	C++
1 ←	1	$1 \times 1 \leq 1000$ true	2	1
2 ←	2	$2 \times 2 \leq 1000$ true	3	2
3 ←	3	$3 \times 3 \leq 1000$ true	4	3
4 ←	4	$4 \times 4 \leq 1000$ true	5	4
⋮				
30 ←	30	$30 \times 30 \leq 1000$ true	31	30
31 ←	31	$31 \times 31 \leq 1000$ true	32	31
	32	$32 \times 32 \leq 1000$ False		

2. constant(int n)

$n = 1000$

C = 0	i	$i < 100$	i++	C++
1 ←	0	true	1	1
2 ←	1	true	2	2
3 ←	3	true	3	3
4 ←	4	true	4	4
5 ←	5	true	5	5
⋮				
100 ←	99	true	100	100
	100	False		